Regular consumption of dark chocolate is associated with low serum concentrations of C-reactive protein in a healthy Italian population.

Dark chocolate contains high concentrations of flavonoids and may have antiinflammatory properties. We evaluated the association of dark chocolate intake with serum C-reactive protein (CRP). The Moli-sani Project is an ongoing cohort study of men and women aged >/=35 y randomly recruited from the general population. By July 2007, 10,994 subjects had been enrolled. Of 4849 subjects apparently free of any chronic disease, 1317 subjects who declared having eaten any chocolate during the past year (mean age 53 +/- 12 y; 51% men) and 824 subjects who ate chocolate regularly in the form of dark chocolate only (50 +/- 10 y; 55% men) were selected. High sensitivity-CRP was measured by an immunoturbidimetric method. The European Prospective Investigation into Cancer and Nutrition FFQ was used to evaluate nutritional intake. After adjustment for age, sex, social status, physical activity, systolic blood pressure, BMI, waist:hip ratio, food groups, and total energy intake, dark chocolate consumption was inversely associated with CRP ($P = 0.038$). When adjusted for nutrient intake, analyses showed similar results ($P = 0.016$). Serum CRP concentrations [geometric mean (95% CI)] univariate concentrations were 1.32 (1.26-1.39 mg/L) in nonconsumers and 1.10 (1.03-1.17 mg/L) in consumers ($P < 0.0001$). A J-shaped relationship between dark chocolate consumption and serum CRP was observed; consumers of up to 1 serving (20 g) of dark chocolate every 3 d had serum CRP concentrations that were significantly lower than nonconsumers or higher consumers. Our findings suggest that regular consumption of small doses of dark chocolate may reduce inflammation.